

D396

Evidence-Based Practice for Health and Human Services

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A community health educator at a county clinic that opened in 2019 wonders whether text message reminders, compared with postcard reminders, raise flu vaccination rates among adults over 65. The clinic has a staff of 14. In PICO terms, what is the “C”?

- A. Text message reminders
- B. Flu vaccination rates
- C. Postcard reminders
- D. A staff of 14

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A PICO question reads: “In adults with low back pain (P), does yoga (I) compared with physical therapy (C) ...” Which ending supplies a proper O?

- A. “... reduce pain scores at 12 weeks?”
- B. “... compared with no treatment?”
- C. “... always cure back pain?”
- D. “... in outpatient clinics?”

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A wellness coordinator finds four sources on workplace walking programs: a blog by a popular fitness coach with 2 million followers, a single randomized trial, a systematic review of 14 randomized trials, and a hospital’s internal report. Which provides the strongest evidence?

- A. The internal report.
- B. The systematic review of 14 randomized trials.
- C. The single randomized trial.
- D. The fitness coach’s blog.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A team finds no systematic review on its question. Which available source should it rank highest?

- A.** An editorial by a national expert panel.
- B.** A case series from a respected hospital.
- C.** A cohort study with 50,000 participants.
- D.** A well-designed randomized controlled trial.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A clinic team has written a PICO question about fall prevention and has searched two databases, finding eight articles. The team lead has a doctorate and 20 years of experience. What is the next EBP step?

- A.** Critically appraise the eight articles for quality and relevance.
- B.** Implement the program from the most recent article.
- C.** Evaluate the outcomes of the program.
- D.** Rely on the lead's experience instead of the articles.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A researcher at a large academic medical center wants to understand how new immigrant mothers experience prenatal care. The center has a \$3 million research budget. Which design best fits?

- A.** An RCT, because the budget can pay for one.
- B.** A meta-analysis.
- C.** A qualitative design using in-depth interviews.
- D.** A randomized controlled trial.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

In a study of a new diabetes education class at three community centers, patients chose whether to attend the class or get usual care. Those who chose the class had better A1c results. The study enrolled 1,200 patients. What is the main weakness?

- A. The study should have used qualitative interviews.
- B. Self-selection: motivated patients may have chosen the class, so the groups differed from the start.
- C. The sample was too small.
- D. A large sample removes all confounding.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A trial of a new smoking cessation app reports a relative risk of relapse of 0.85 (95% CI 0.70 to 1.05). The app won a design award and has 4.5 stars in app stores. How should the result be read?

- A. A significant 15% reduction in relapse.
- B. Significant, because the interval does not include 0.
- C. Proven effective, given its high rating.
- D. Not statistically significant, because the interval includes 1.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

Which result for a mean difference in blood pressure is statistically significant at the 95% level?

- A. -3 mmHg (95% CI -5 to 2)
- B. Any result whose interval excludes 1.
- C. -4 mmHg (95% CI -7 to -1)
- D. -6 mmHg (95% CI -13 to 1)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

The best evidence favors a supervised exercise program for a client's knee arthritis. The client, a 60-year-old widower who owns a hardware store, says he will not attend group classes but would walk at home. What is the evidence-based response?

- A.** Suggest he sell the store to free up time.
- B.** Drop exercise entirely since he refused the class.
- C.** Insist on the group program because the evidence favors it.
- D.** Adapt the plan to a home walking program with supports, respecting his preference.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

In a trial, 20% of control patients and 12% of patients using a new fall-prevention program fell within a year. The program runs in 30 senior centers. How many patients need the program to prevent one fall?

- A.** 13
- B.** 8
- C.** 5
- D.** 30

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

An intervention has an NNT of 25. What is its absolute risk reduction?

- A.** 96%
- B.** 4%
- C.** 25%
- D.** 0.4%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A community health educator at a county clinic ~~that opened in 2019~~ wonders whether text message reminders, compared with postcard reminders, raise flu vaccination rates among adults over 65. ~~The clinic has a staff of 14.~~ In PICO terms, what is the “C”?

Target word: identify comparison **Decides:** The alternative the intervention is compared against.

A	Confusion set	That is the Intervention.
B	Confusion set	That is the Outcome.
C	Correct	The comparison is the alternative to the intervention.
D	Camouflage	Setting detail, not a PICO element.

2. Answer A Concept 1 • transfer

A PICO question reads: “In adults with low back pain (P), does yoga (I) compared with physical therapy (C) ...” Which ending supplies a proper O?

Target word: reverse: complete the missing element **Decides:** The outcome is a measurable result.

A	Correct	A measurable outcome with a time frame.
B	Confusion set	A second comparison, not an outcome.
C	Absolute language	Absolute and not measurable as written.
D	Adjacent category	A setting, not an outcome.

3. Answer B Concept 2

A wellness coordinator finds four sources on workplace walking programs: a blog ~~by a popular fitness coach with 2 million followers~~, a single randomized trial, a systematic review of 14 randomized trials, and a hospital’s internal report. Which provides the strongest evidence?

Target word: top of the hierarchy **Decides:** Systematic review of RCTs.

A	Adjacent category	A quality improvement report ranks low.
B	Correct	Pools multiple RCTs; highest level.
C	Dropped element	Strong, but one trial ranks below a review of many.
D	Camouflage	Popularity is not evidence.

4. Answer D Concept 2 • transfer

A team finds no systematic review on its question. Which available source should it rank highest?

Target word: reverse: next level down **Decides:** Below systematic reviews come individual RCTs.

A	Inverted operation	Expert opinion sits at the bottom of the hierarchy.
B	Adjacent category	Case series rank below both RCTs and cohort studies.
C	Absolute language	Size does not lift an observational design above an RCT.
D	Correct	Single RCTs are the next level below systematic reviews.

5. Answer A Concept 3

A clinic team has written a PICO question about fall prevention and has searched two databases, finding eight articles. ~~The team lead has a doctorate and 20 years of experience.~~ What is the next EBP step?

Target word: next step **Decides:** After acquire comes appraise.

A	Correct	Appraise before applying.
B	Dropped element	Skips appraisal.
C	Inverted operation	Assessing outcomes is the last step.
D	Camouflage	Experience matters but does not replace appraisal.

6. Answer C Concept 4

A researcher at a large academic medical center wants to understand how new immigrant mothers experience prenatal care. ~~The center has a \$3-million research budget.~~ Which design best fits?

Target word: design for question type **Decides:** "Experience of" is a qualitative question.

A	Camouflage	Budget does not decide the design.
B	Adjacent category	Pools quantitative studies; not a primary design for this question.
C	Correct	Explores meaning and experience.
D	Confusion set	Tests effects, not experiences.

7. Answer B Concept 5

In a study of a new diabetes education class at three community centers, patients chose whether to attend the class or get usual care. Those who chose the class had better A1c results. ~~The study enrolled 1,200 patients.~~ What is the main weakness?

Target word: source of bias **Decides:** Self-selection — no randomization.

A	Adjacent category	The question is about effect, so design fit was fine.
B	Correct	Without randomization, confounders are not balanced.
C	Camouflage	1,200 is large; size does not fix selection bias.
D	Absolute language	Size does not replace random assignment.

8. Answer D Concept 6

A trial of a new smoking cessation app reports a relative risk of relapse of 0.85 (95% CI 0.70 to 1.05). ~~The app won a design award and has 4.5 stars in app stores.~~ How should the result be read?

Target word: significance from CI **Decides:** CI for a ratio includes 1.

A	Dropped element	Reads the point estimate and ignores the interval.
B	Inverted operation	0 is the null for differences; for ratios it is 1.
C	Camouflage	Ratings are not evidence.
D	Correct	A ratio CI that crosses 1 is compatible with no effect.

9. Answer C Concept 6 • transfer

Which result for a mean difference in blood pressure is statistically significant at the 95% level?

Target word: reverse: pick the significant interval **Decides:** Difference CI must exclude 0.

A	Confusion set	Includes 0.
B	Inverted operation	Applies the ratio rule to a difference; for differences the null value is 0.
C	Correct	The interval does not include 0.
D	Dropped element	Larger point estimate, but the interval includes 0.

10. Answer D Concept 7

The best evidence favors a supervised exercise program for a client's knee arthritis. The client, a 60-year-old widower ~~who owns a hardware store~~, says he will not attend group classes but would walk at home. What is the evidence-based response?

Target word: integrating preferences **Decides:** Client preference is one of the three parts of EBP.

A	Camouflage	The business detail is irrelevant.
B	Inverted operation	Swings to abandoning the evidence.
C	Absolute language	Ignores client values.
D	Correct	EBP blends evidence, expertise, and client values.

11. Answer A Concept 8

In a trial, 20% of control patients and 12% of patients using a new fall-prevention program fell within a year. ~~The program runs in 30 senior centers.~~ How many patients need the program to prevent one fall?

Target word: compute NNT **Decides:** $ARR = 20\% - 12\% = 8\%$.

A	Correct	$1 / 0.08 = 12.5$, rounded up to 13.
B	Inverted operation	The ARR in percent, not its inverse.
C	Dropped element	$1 / 0.20$ uses only the control rate.
D	Camouflage	The number of centers.

12. Answer B Concept 8 • transfer

An intervention has an NNT of 25. What is its absolute risk reduction?

Target word: solve for ARR **Decides:** $ARR = 1 / NNT$.

A	Inverted operation	$1 - 0.04$ — the share not helped.
B	Correct	$1 / 25 = 0.04$.
C	Wrong stem number	Reads the NNT as a percentage.
D	Confusion set	Misplaces the decimal: $1/25$ is 0.04, which is 4%.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	8	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Building a PICO(T) question	A searchable clinical question names the Population, Intervention, Comparison, Outcome, and sometimes Time.	Mixing the intervention and the outcome, or leaving out the comparison.	A practice question to rewrite or a PICO part to identify.
2	Levels of evidence	Systematic reviews and meta-analyses of randomized trials sit at the top, then single RCTs, then cohort and case-control studies, then case reports and expert opinion.	Ranking a single large study or an expert panel above a systematic review.	Several sources on the same question; which is strongest?
3	The EBP process in order	Ask a focused question, Acquire the evidence, Appraise it, Apply it with expertise and preferences, then Assess the outcome.	Implementing a change before appraising the evidence, or skipping evaluation after rollout.	“What is the next step?” in an EBP project.
4	Design matches the question	Questions about effect or cause fit experimental or cohort designs; questions about experiences and meaning fit qualitative designs.	Using a randomized trial to learn how patients feel, or interviews to prove an effect.	The wording of the research question: “effect of” vs. “experience of.”
5	Randomization and bias	Random assignment balances known and unknown confounders between groups; blinding reduces measurement and performance bias.	Believing a large sample alone removes confounding.	A study that let participants or staff choose group assignment.
6	Statistical vs. practical significance	A 95% confidence interval for a ratio that includes 1 (or for a difference that includes 0) is not statistically significant; a significant result can still be too small to matter in practice.	Treating any reduction in the point estimate as proof of effect.	A ratio with its confidence interval.
7	Three parts of evidence-based practice	EBP integrates the best research evidence, the professional’s expertise, and the client’s values and preferences.	Thinking EBP means applying research findings no matter what the client wants.	A client who declines the option the evidence favors.
8	Number needed to treat	$NNT = 1 / \text{absolute risk reduction}$, where ARR is the control event rate minus the treatment event rate; round up.	Using the relative risk reduction or dividing by one group’s rate.	Two event rates, one per group.

Course scope (public description): Evidence-Based Practice for Health and Human Services prepares the learner to apply evidence-based practice (EBP) to inform healthcare recommendations in out-patient, organizational, and other public health settings. Learners will be introduced to an EBP framework to guide them through the steps of EBP using real world scenarios. There are no prerequisites for this course.